

SINETHEMBA MVELASE

Quantitative Analyst | Machine Learning & Data Science | Banking & Financial Services

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PROFILE SUMMARY

Analytical, technically versatile **Quantitative Analyst** with a **BSc in Mathematics and Physics** and applied experience in **machine learning, data science, statistical modelling, cloud analytics, and executive reporting** across a large financial services group.

Combines a rigorous mathematical foundation (probability, statistics, linear algebra, mathematical modelling) with production-grade tooling in **Python, SQL, AWS, Google Cloud Platform, BigQuery, Vertex AI, and Power BI** to extract, transform, and model large datasets and translate the results into decisions for senior stakeholders.

Currently a **Discover Graduate Trainee at Vodacom Financial Services** supporting Risk Management, Technology Assurance, Service Management, and AWS Cost Optimization — while independently building AI/ML products (**KasiKash, CAPS AI Tutor**) that apply predictive modelling, alternative credit scoring, and NLP at scale. Positioned to contribute immediately to a **banking quantitative analytics team** focused on **predictive modelling, model development and validation, risk analytics, and data-driven decision science**.

CORE COMPETENCIES

Machine Learning: Supervised & Unsupervised Learning • Regression • Classification • Decision Trees • Random Forest • Clustering • Feature Engineering • Model Evaluation • Predictive Modelling • Statistical Modelling • Model Monitoring
Statistics & Mathematics: Probability • Statistical Analysis • Hypothesis Testing • Linear Algebra • Applied Mathematics • Mathematical Modelling • Quantitative Analysis
Data Science & Engineering: Data Cleaning • Data Wrangling • Data Transformation • Exploratory Data Analysis (EDA) • Feature Engineering • Big Data Handling • ETL Workflows
Cloud & Databases: AWS • AWS Bedrock • Google Cloud Platform (GCP) • BigQuery • Vertex AI • Supabase • PostgreSQL
Reporting, BI & Analytics: Power BI • Advanced Excel • PowerPoint • Dashboard Development • Executive Reporting • KPI Analytics • Business Intelligence • Data Visualisation
Programming & DevOps: Python (pandas, NumPy, scikit-learn) • SQL • TypeScript • JavaScript • Git • GitHub Actions • CI/CD • Vercel • Render
Domain & Governance: Financial Services Analytics • Risk Analytics • Model Development • Model Validation • Governance Reporting • ITIL 4

TECHNICAL SKILLS MATRIX

Domain	Tools & Techniques	Proficiency
Machine Learning	Regression, Classification, Random Forest, Clustering, Model Evaluation	Intermediate–Advanced
Statistics & Mathematics	Probability, Hypothesis Testing, Linear Algebra, Mathematical Modelling	Advanced
Python for Data Science	pandas, NumPy, scikit-learn, matplotlib, seaborn	Advanced
SQL & Databases	BigQuery SQL, PostgreSQL, Supabase	Advanced
Cloud Platforms	AWS, AWS Bedrock, GCP, Vertex AI	Intermediate–Advanced
Business Intelligence	Power BI, Excel (Advanced), PowerPoint, Dashboards	Advanced
Data Engineering	Data Cleaning, Wrangling, Transformation, EDA	Advanced
DevOps / MLOps	Git, GitHub Actions, CI/CD, Vercel, Render	Intermediate
Financial Analytics	Executive Reporting, KPI Analytics, Cost & Risk Reporting	Intermediate

PROFESSIONAL EXPERIENCE

Discover Graduate Trainee — Technology Management Feb 2025 – Present
Vodacom Financial Services — Johannesburg, South Africa

Embedded within senior technology leadership to deliver quantitative analytics, executive reporting, and governance support across Risk Management, Technology Assurance, Service Management, PI Delivery, and AWS Cost Optimization.

- Deliver **quantitative analysis, trend reporting, and executive insights** for senior management, presented at **Steering and Management Committees**.
- Design **analytical presentations, performance reports, and dashboards** used by ExCo-level stakeholders for strategic decision-making across the Financial Services division.
- Support **Risk Management** and **Technology Assurance** with data extraction, KPI tracking, and reconciliation of metrics — foundational to **model development, model monitoring, and model validation** work.
- Contribute to **AWS Cost Optimization**, analysing cloud consumption datasets, identifying variance drivers, and producing optimisation recommendations.
- **Automate reporting and analytical workflows** with Python and advanced Excel, improving speed, accuracy, and data-quality controls.
- Collaborate with executives and cross-functional stakeholders (Finance, Technology, Operations, Assurance) to translate business questions into data problems.

Reporting Analyst Intern Jun 2024 – Jan 2025
Vodacom Financial Services — Johannesburg, South Africa

- Developed and maintained **Power BI dashboards** and **advanced Excel models** tracking KPIs and business performance for internal stakeholders.
- Performed **data extraction, cleaning, and transformation** across multiple sources to ensure reporting accuracy and consistency.
- Delivered recurring and ad-hoc **business performance analytics**, converting raw data into commercial insight.
- Automated repetitive reporting processes using Excel and Power BI, reducing turnaround time and error rates.

Vodafone Portugal — Data Science Project

- Extracted large-scale enterprise datasets from **Google BigQuery** for downstream modelling and analytics.
- Performed **data cleansing, preprocessing, transformation, and exploratory data analysis (EDA)** on Google Cloud Platform.
- Applied **machine learning and statistical techniques** (regression, classification, clustering, model evaluation) using **Vertex AI** to derive business insights.
- Delivered analytical outputs and recommendations informing data-driven decisions for the Vodafone Portugal team.

PROJECTS

KasiKash — AI-Powered Financial OS with Alternative Credit Scoring

Founder & Lead Developer

- Developed a **voice-first, multilingual financial management platform** for township SMEs, converting spoken transactions into structured bookkeeping records via **NLP** (English, isiZulu, Sesotho).
- Designed "**KasiScore**", a proprietary **alternative credit-scoring model** built on transaction volume, frequency, repayment behaviour, and stokvel contribution patterns — a hands-on **predictive modelling** exercise directly analogous to **credit risk analytics** in banking.
- Implemented **transaction classification and financial analytics** features (feature engineering, categorisation, reconciliation).
- **Stack:** React, TypeScript, Supabase, PostgreSQL, GitHub Actions CI/CD, deployed on Vercel.

CAPS AI Tutor — Conversational AI Learning Platform

Founder & AI Engineer

- Built a conversational **LLM-powered tutor** aligned to the South African **CAPS curriculum**, deployed on **AWS Bedrock** with **Anthropic Claude**.
- Integrated **WhatsApp** and **USSD** channels for low-bandwidth access; implemented multilingual NLP support.
- Deployed on **Vercel** and **Render** with **GitHub Actions CI/CD**, applying MLOps practices for prompt versioning and cost control.

Enterprise Delivery Management Platform

Full-Stack Engineer

- Designed the **relational data model** and **PostgreSQL schema** underpinning driver trip tracking, cash-on-delivery reconciliation, and dispatch analytics.
- Built **real-time operational analytics** using Node.js, Prisma, Redis, and Socket.IO.
- Implemented secure authentication (JWT) and automated workflows; a practical exercise in translating **business processes into data models**.

EDUCATION

Enterprise and Technology Management

NWU Business School • 2024

BSc Mathematics and Physics

University of Johannesburg • 2019 – 2023

Key modules: *Applied Mathematics, Probability, Statistics, Linear Algebra, Mathematical Modelling, Numerical Methods.*

KEY STRENGTHS

- Rigorous quantitative foundation (BSc Mathematics & Physics) applied to real financial services data.
- Fluent in **Python and SQL** for data engineering, feature engineering, statistical analysis, and ML.
- Practical **predictive modelling** experience via **KasiScore** — an alternative credit-scoring engine.
- Delivering **executive analytics and dashboards** in a regulated financial services environment.
- Cloud-native ML workflows on **AWS, GCP, BigQuery, and Vertex AI**.

ATS KEYWORDS

Machine Learning • Data Science • Python • SQL • Statistical Analysis • Feature Engineering • Predictive Modelling • Classification Models • Regression Models • Clustering • Model Development • Model Validation • Model Monitoring • Data Analytics • Big Data • Data Wrangling • Business Intelligence • Financial Services • Banking Analytics • Cloud Computing • AWS • AWS Bedrock • GCP • BigQuery • Vertex AI • Power BI • Quantitative Analysis • Risk Analytics • Data Visualisation • Executive Reporting

CERTIFICATIONS

Machine Learning Specialization

DeepLearning.AI / Stanford University (Andrew Ng)

ITIL 4 Foundation

IT Service Management

LANGUAGES

- English — Professional
- isiZulu — Native
- Sesotho — Conversational

REFERENCES

Available on request.